Prevalence of Oral Candidiasis in both Diabetic and Non diabetic Subjects of Andhra Pradesh: a Pilot Study

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Prevalence of Oral Candidiasis in both Diabetic and Non diabetic Subjects of Andhra Pradesh: a Pilot Study

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ABSTRACT: Candidiasis is emerged as an important opportunistic or secondary infection in some metabolic disorders like diabetic, immunocompromised and immunosuppressed patients resulting in significant morbidity and mortality. The Present study was aimed to determine the prevalence of oral candidiasis in both diabetic and non-diabetic subjects of Andhra Pradesh. Total 100 subjects were enrolled in the study (includes diabetic and non-diabetic) and oral washing were collected and analyzed by using various phenotypic and genotypic methods. Oral Candidiasis was found to be significantly higher in diabetic subjects (85.3%) when compare to the healthy individuals (14.7%) and moreover positive correlation was observed in between hyper glycemic levels and levels of candida colonization. Further we observed that C. albicans was the most prevalent species in both diabetics (81.8%) and healthy individuals (18.2%). Candida non-albicans which includes C. tropical is, C. parapsilosis, C. ontarioensis and Dipodascusis capitates was isolated only from diabetic patients. Our findings were also recorded thatthe oral Candidiasis was significantly higher in smokers than non-smokers in both diabetic and healthy control.

KEYWORDS: Oral candidiasis, Diabetic patients, Candida albicans, Candida non-albicans, Nosocomial infections

1. INTRODUCTION

Candida albicans (C. albicans) is the most common fungal species that is isolated from the oral cavity of healthy individuals [1-3]. It has been estimated that oral Candida species harbor the oral cavity of 17% to 75% systemically healthy individuals [4]. However, under opportunistic conditions, these fungi may become opportunistic pathogens [3]. Local risk factors have been associated with an increased oral Candida prevalence and carriage includes xerostomia, tobacco smoking, denture wearing and poor oral hygiene maintenance [5-8]. Systemic diseases such as poorly-controlled diabetes mellitus, acquired immune deficiency syndrome and renal disorders have also been associated with an increased oral Candida carriage, which make immunosuppressed patients more susceptible to develop oral candidiasis as compared with their systemically healthy counterparts [3,9-11].Studies [3,12,13] have reported that oral Candida carriage is significantly higher in patients with chronic hyperglycemia (such as those with poorly-controlled diabetes) than non-diabetic controls. These results may be explained by the fact that a dry oral environment (due to xerostomia in patients with poorly-controlled diabetes) facilitates Candida stagnation and growth on oral surfaces, most commonly the dorsal surface of tongue [3,14]. Moreover an immunocompromised state in patients with chronic hyperglycemia may also facilitate oral Candida growth and proliferation.

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The precise mechanism through which oral Candida carriage is affected by tobacco remains unclear. However, it has also been hypothesized the aromatic hydrocarbons (such as N-nitrosobenzylmethylamine) in tobacco smoke act as nutrients for Candida species thereby augmenting their growth [15]. Moreover, it has also been reported that C. albicans catalyzes the formation of N-nitrosobenzylmethylamine, which is an explanation for the high prevalence of Candida-associated leukoplakia among tobacco smokers as compared with non-smokers [16].

Since chronic hyperglycemia and tobacco smoking are independent risk factors for increased oral Candida carriage; it is hypothesized that oral Candida carriage and species prevalence is significantly higher in type 2 diabetic and non-diabetic smokers as compared to non-smokers with and without diabetes. The aim of the present study was to assess the oral Candida carriage and species prevalence among tobacco smokers and non-smokers with and without diabetes.

2. MATERIALS AND METHODS

Study population and Ethical issues:

Patients who were attending at diabetic outpatient department of Rajiv Gandhi Institute of Medical Sciences hospital, kadapa randomly selected, and the same number of healthy individuals was also enrolled in the present study. The present study protocols were approved by the Research Ethics Review committee at the College of Yogi Vemana University, Kadapa, Andhra Pradesh. Participation was voluntary and all participants were asked to sign on consent form before being included in the present investigation.

Oral rinsing:
Oral yeast colonization was assessed with the concentrated oral rinse technique as described by Samaranayake et al [17] was used to collect samples. Briefly, subject in both the groups were instructed not to eat and drink 2 h prior to the sample collection. Each individual was given 10 ml of sterile phosphate buffer saline solution (0.1 M pH 7.2) and was asked to rinse mouth for 60 s with this solution. Then that mouth rinse was collected in sterile containers and was transported to the microbiological laboratory.

Isolation and identification of Candida species:
As soon as we received the oral rinsing samples at the laboratory 100µl of the sample was inoculated on sabourad dextrose agar plates and incubated at 27°C for 24hr. After completion of incubation period distant colonies were observed and selected some individual prominent colonies were selected to identify by microscopic observation, Cultural techniques and molecular techniques.C.albicans and C.tropicalis was differentiated from the other Candida species by germ tube method. Selected Candida colonies were inoculated into the plasma and incubated at 37°C for 3hr. Then the aliquots were subjected for microscopic observation. Germ tube was observed as a slender tube with straight walls without septum and without constriction at the junction between the cells.We used the CHROM agar medium which is a selective and differential medium for the identification of dimorphic fungi with the inclusion of chromogenic substrates in the medium. The main advantage of this medium is, we can easily differentiate and detect mixed yeast population depending on the color they produced. Distinguished selected colonies of Candida sps were inoculated on CHROM agar plates to identify at the species level depending on the different colors they produced. Usually Candida albicans appear light green in color, Candida tropicalis appear bluish green and light pink in color, creamy etc other yeast may develop either natural color or appear rose light to dark.The selected Candida sps from the CHROM agar plates were further analyzed by using various molecular methods i.e. by amplifying fragment of 18S rRNA by using PCR and also applied several bioinformatics tools i.e. nucleotide sequence homology and phylogenetic analysis were used to identify the different Candida species.Statistical analysis Group analyses were performed with the chi-square test and the differences between two independent samples were analyzed with the Student's t-test.

3. RESULTS

Characteristics of the study population:
The mean age of diabetic groups was 26.8 years, and 35% of the participants were male and 15 % were females. Among the male population 16 (22.8%) smokers and remaining were nonsmokers. The mean age of non-diabetic groups was 28.6 years, and 40% of the participants were male and 10 % were females. Among the male population 18 (22.5%) smokers and remaining were nonsmokers.

Oral Candida carriage and species prevalence:
In total, 40% (8/20) of the diabetic patients and 15% (3/20) of the controls yielded Candida organisms on culture. The mean counts of Candida colonies in diabetic and control groups were 825.25 CFU/mL and 205.7CFU/mL, respectively. However, the difference between the two groups was not statistically significant (t=5.6030; P=0.0053). Gender had no influence on the yeast carriage in the mouth. The habit of betel chewing and clinical staging had no influence on the Candidal carriage status of OSMF patients. Cytological buccal smears detected no yeast cells or pseudo hyphae in either group.

C. albicans was the most commonly isolated yeast species among diabetic patients and controls. C. albicans carriage was significantly higher in diabetic smokers as compared with non-diabetic smokers. Among patients without diabetes, C. albicans carriage (50.2%) was significantly higher as compared with healthy individuals. In 81.7% non-diabetic control group no yeast species were identified. None of the study participants harbored other oral Candida species (Candida krusei, Candida glabrata, Candida luscitanie and Candida gullerimondi).

In the present study we also isolated and identified Candida non-albicans species from the diabetic group only. Among the Candida non-albicans C.tropicalis, C. parapsilosis C. ontarioensis and Dipodascuscapitatus were isolated only from diabetic populations but not in healthy individuals. The prevalence of Candida non-albicans in diabetic population was observes as (15%) C. tropicalis, (20%) of C. parapsilosis, (15%) C.ontarioensis and (10%) Dipodascuscapitatus. We also observed a positive correlation in between the levels of hyper glycemia and the no of CFU of the isolate. In the present study we could not isolated any of the Candida non-albicans species from the healthy individuals.

As shown in the (Fig 2) germ tube test is the predominant and preliminary test for the detection of the C.albicans and C.tropicalis, further, all the isolated Candida sps from SDA were selected and re-inoculated on the CHROM agar medium for identification by color which they roduced. 40 isolates were chosen for present study and very interestingly, almost all the species were showed magic in the color formation that is green color by C.albicans, the

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distinctive dark blue-gray color shown by C.tropicalis, only dark gray color by C.ontarioensis and pink-white mixed margins by Dipodascusiscapitates(Table 1).

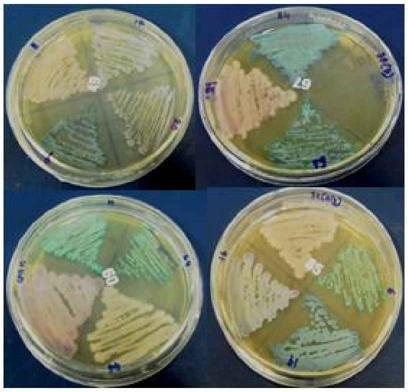

Fig: 1: Identification of different candida species on CHROM agar depending on the colony color.

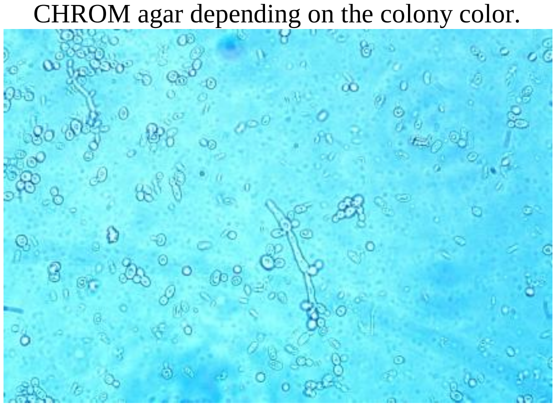

Fig: 2 : Germ tube method to differentiate C.albicans or C.tropicalis

Table 1: The detailed information of the patients and health status included in the present study

The species which were differentiated by the CHROM agar were subjected to PCR amplification and electrophoresis, for further identification of species. Based on the differences in bands formation on the agarose gel electrophoresis the different species were selected for further sequencing process. (Fig 4). The 18S region was sequenced and BLAST analysis was performed with NCBI Database of gene bank. Based on the maximum identity score all the species were named and their Gene Bank accession numbers were recorded. Candida species which is showing the pink color colonies and the blue color colonies on the CHROM agar showed 99% of similarity with the Candida tropicalis (JQ008834.1) and identifies as Candida tropicalis. Another Candida species which is showing the grayish purple on the CHROM agar and has 99% of similarity with the Candida ontarioensis (JN820129.1) and was identified as Candida ontarioensis. The Candida species showing the green color colonies on the CHROM agar and has maximum (99%) of similarity with the Candida albicans (HQ876034.1) was identified as Candida albicans. The colonies showing the pink colonies with the white borders on the CHROM agar and showing 99% of similarity with the gene bank accession number ab083080.1 and identified as Dipododascuscapitatus.

S:no	Name of the patient	Sex	Age	Colony Colour on CHROM agar	Germ tube test	Isolate	Signs of infection	Habits
1	Subject-1	M	30	Green	+ve	*Candida albicans*	symptomatic	-
2	Subject-2	F	60	Green	+ve	*Candida albicans*	symptomatic	-
3	Subject-3	M	50	Green	+ve	*Candida albicans*	symptomatic	smoking
4	Subject-4	M	60	Green	+ve	*Candida albicans*	symptomatic	smoking
5	Subject-5	M	60	Green	+ve	*Candida albicans*	symptomatic	-
6	Subject-6	F	55	Green	+ve	*Candida albicans*	asymptomatic	-
7	Subject-7	M	70	Green	+ve	*Candida albicans*	asymptomatic	smoking
8	Subject-8	F	41	Green	+ve	*Candida albicans*	asymptomatic	-
9	Subject-9	F	56	Green	+ve	*Candida albicans*	asymptomatic	-
10	Subject-10	M	68	Green	+ve	*Candida albicans*	asymptomatic	-
11	Subject-11	M	40	Green	+ve	*Candida albicans*	asymptomatic	smoking
12	Subject-12	M	35	Green	+ve	*Candida albicans*	asymptomatic	-
13	Subject-13	F	44	Green	+ve	*Candida albicans*	asymptomatic	-
14	Subject-14	F	50	Green	+ve	*Candida albicans*	asymptomatic	-
15	Subject-15	M	45	Green	+ve	*Candida albicans*	asymptomatic	Smoking
16	Subject-16	M	43	Green	+ve	*Candida albicans*	asymptomatic	-
17	Subject-17	M	55	Green	+ve	*Candida albicans*	asymptomatic	Smoking

Further the infection caused by the Candida sps is compared with several factors, the range of infection in the symptomatic and asymptomatic oral candidiosis can be assessed by the no: of colonies formed on the SDA plate. The plate which is inoculated with symptomatic infection sample shown more number of colonies than the asymptomatic samples, by this range of infection can be assessed, graph (Fig-3) is plotted by comparing five symptomatic infectious patients (white patchy appearance on the buccal cavity) and asymptomatic infectious patients (samples were collected from the diabetic patients with no oral infection symptoms) the meanvalue of the no: of colonies is represented. Another factor is age that increases the infection with the increase in the age, samples were collected from different age groups i.e. from 30-70 years, and these were compared to controls, the range of infection is measured by taking mean values of no: of colonies formed that is plotted as a graph (Fig-4).

18	Subject-18	F	60	Green	+ve	*Candida albicans*	asymptomatic	-
19	Subject-19	M	70	Green	+ve	*Candida albicans*	asymptomatic	-
20	Subject-20	M	60	Green	+ve	*Candida albicans*	asymptomatic	-
21	Subject-21	M	75	Green	+ve	*Candida albicans*	asymptomatic	-
22	Subject-22	F	46	Green	+ve	*Candida albicans*	asymptomatic	-
23	Subject-23	F	33	Baby Pink	+ve	*Candida tropicalis*	asymptomatic	-
24	Subject-24	M	56	Baby pink	+ve	*Candida tropicalis*	asymptomatic	-
25	Subject-25	M	62	Baby pink	+ve	*Candida tropicalis*	asymptomatic	smoking
26	Subject-26	M	43	Metallic blue	+ve	*Candida tropicalis*	asymptomatic	-
27	Subject-27	M	60	Metallic blue	+ve	*Candida tropicalis*	asymptomatic	smoking
28	Subject-28	F	60	Metallic blue	+ve	*Candida tropicalis*	asymptomatic	-
29	Subject-29	M	50	Metallic blue	+ve	*Candida tropicalis*	asymptomatic	-
30	Subject-30	M	42	Metallic blue	+ve	*Candida tropicalis*	asymptomatic	smoking
31	Subject-31	M	46	Metallic blue	+ve	*Candida tropicalis*	asymptomatic	Smoking
32	Subject-32	M	55	Metallic blue	+ve	*Candida tropicalis*	asymptomatic	smoking
33	Subject-33	M	58	Cream colour	-ve	*Candida parapsilosis*	asymptomatic	Smoking
34	Subject-34	F	60	Cream colour	-ve	*Candida parapsilosis*	asymptomatic	-
35	Subject-35	M	39	Cream colour	-ve	*Candida parapsilosis*	asymptomatic	-
36	Subject-36	F	43	Grayish purple	-ve	*Candida ontarioensis*	asymptomatic	-
37	Subject-37	M	46	Grayish purple	-ve	*Candida ontarioensis*	asymptomatic	-
38	Subject-38	M	43	Grayish purple	-ve	*Candida ontarioensis*	asymptomatic	-
39	Subject-39	F	66	Pink with white borders	-ve	*Dipodascus capitatus*	asymptomatic	-
40	Subject-40	F	50	Pink with white borders	-ve	*Dipodascus capitatus*	asymptomatic	-

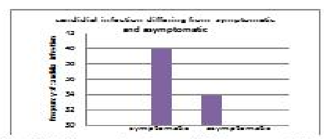

Fig 3: Graphical representation of range of infection symptomatically and asymptomatically.

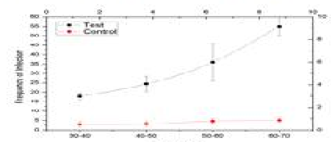

Fig 4: Graphical representation of range of infection among smokers and non-smokers.

4. DISCUSSION

The present study has confirmed that Candida is more prevalent in the oral cavity of diabetics than non-diabetics however the carriage rate of Candida in the oral cavity was differentin various studies. The oral rinse technique with PBS was used for sampling in the present study since this is known to be a sensitive technique for estimating the oral Candidal carriage (17, 18) there was a significant difference in the carriage rate of Candida in the oral cavity of diabetics in various aspects. Colony forming unit (CFU) is usually recorded to obtain the clinical data essentially to establish a clinical diagnosis of oral candidiasis. (19) Epstein et al (1980) have demonstrated that carriers and patients with oral candidiasis can be reliably distinguished on the basis of quantitative cultures. The patients with clinical candidiasis harbor large number of colonies on SDA than the patients who are the carriers i.e. (asymptomatic) such cut off limits for CFU between carriers and patients may serve as a useful clinical indicator. It is observed in the present study from the (Fig-3). The rate of infection was calculated by the number of colonies formed on SDA of different age group samples, there is no significant range of infection of the samples below 30years, inspite that higher rate of carriage in age group 41-70, Tapper-Jones et al (1981) (20), while Smits B.J.et al (1996) (21) has been proved significantly that the increased prevalence of Candida albicans advancing with different age groups and also with support of other predisposing factors (Fig-4).

Various mechanisms were identified for Candida carriage with respect to habits of the diabetic subject that successively effect on the metabolic disorders by smoking, alcoholic addiction but still it is uncertain. Arendrof and Walker (1980) (22) were proposed that smoking may lead to localized epithelial alterations which facilitate colonization of Candida sps which regulates in subclinical infection. In addition to diabetes mellitus, the prevalence of oral Candida infections was influenced primarily by smoking Abu-Elteen KH (23), Neville B (24), Arendorf TM, (1983) (25), Rindum JL (1994) (26). Present study clearly stating that extensive cases of Candida infections were facilitated that the findings of infected individuals specifically non- smokers of diabetic subjects are not or less prone to C. albicans colonization. The range of infection between non- smokers and smokers are assessed by the growth, number of colonies formed on SDA medium and also represented by the appearance of the colony and its characterization, and remarkably their significant levels were shown in (fig-4).

5. CONCLUSION

The present investigation has been however disclosed several local factors that ultimately influencing the rate of Candida infection with reference to the age, habits (smoking, alcoholic addiction etc.,), symptomatic and asymptomatic predisposing factors. Special attention over these findings that the local factors were reducing the high risk of thrush development or oral colonization levels of Candida infection in diabetic patients.

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